Financial Analysis Using Excel

Project Summary

This project demonstrates the application of financial analysis techniques using Microsoft Excel. Various financial functions and analytical methods such as annuity valuation, EMI calculation, investment decision analysis, Net Present Value (NPV), Internal Rate of Return (IRR), Extended Internal Rate of Return (XIRR), and Modified Internal Rate of Return (MIRR) were implemented. The objective of the project was to analyze different financial scenarios and evaluate investment opportunities using discounted cash flow techniques. The results provide insights into loan repayment structures, investment viability, and project comparison.

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1. Annuity Calculation

An annuity represents a series of equal payments made at regular intervals. In this section, the present value of an annuity was calculated using the payment amount, interest rate, and number of payment periods.

The inputs included:

- Price: 32,000
- Interest Rate: 13%
- Number of Payments: 8
- Payment per Period: -6000

Using the present value annuity formula in Excel, the present value of the annuity was calculated as **₹28,793**.

Annuity				
Price	32000		Price	32000
Interest Rate	0.13		Interest Rate	0.13
No. of payments	8		No. of payments	8
Payment	-6000		Payment	-6000
Payments at end of the year			Payments at end of the year	
PV	₹ 28,793		PV	₹ 28,793

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2. EMI Calculation

Equated Monthly Instalment (EMI) is used to determine the fixed monthly payment required to repay a loan over a specific period. In this section, EMI was calculated using loan amount, interest rate, and loan tenure.

Key inputs:

- Annual Interest Rate: 12%
- Monthly Interest Rate: 1%
- Loan Term: 25 years
- Loan Amount: ₹5,000,000

Using Excel financial formulas, the EMI was calculated as **₹52,661.21**.

EMI

Rate per annum	0.12
Rate per month	0.01
Term	25
No. of monthly payment	300
Loan Amount (PV)	-5000000
FV	0
Type	1
EMI	₹ 52,661.21

Rate per annum	0.16
Rate per month	0.013333333
Term	8
Loan Amount (PV)	-100000
FV	0
Type	0
EMI	₹ 13,261.59

Rate per month	0.013
No. of monthly paymen	8
Loan Amount (PV)	100000
FV	0
Type	0
EMI	₹ 13,242.27

Interest paid between 2nd and 3rd months
2132.23

Principal paid between 2nd and 3rd months
24352.30

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3. Loan Amortization Schedule

An amortization schedule was created to analyze how each EMI payment is divided between principal repayment and interest payment over time. The schedule shows the beginning balance, interest amount, principal repayment, and ending balance for each period.

The analysis revealed that during the early periods of the loan, a larger portion of the payment goes toward interest, while the principal repayment gradually increases over time.

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4. Term Loan Analysis

A term loan scenario was analysed to determine the number of payments required to repay the loan.

Inputs:

- Loan Amount: ₹100,000
- Interest Rate: 10%
- EMI Payment: ₹15,000

Using Excel calculations, the loan repayment period was determined to be **7 months**.

Month	Beginning Balance	EMI	Interest	Principal	Ending Balance
1	₹ 1,00,000.00	₹ 13,242.27	₹ 1,300.00	₹ 11,942.27	₹ 88,057.73
2	₹ 88,057.73	₹ 13,242.27	₹ 1,144.75	₹ 12,097.52	₹ 75,960.22
3	₹ 75,960.22	₹ 13,242.27	₹ 987.48	₹ 12,254.78	₹ 63,705.43
4	₹ 63,705.43	₹ 13,242.27	₹ 828.17	₹ 12,414.10	₹ 51,291.34
5	₹ 51,291.34	₹ 13,242.27	₹ 666.79	₹ 12,575.48	₹ 38,715.86
6	₹ 38,715.86	₹ 13,242.27	₹ 503.31	₹ 12,738.96	₹ 25,976.89
7	₹ 25,976.89	₹ 13,242.27	₹ 337.70	₹ 12,904.57	₹ 13,072.33
8	₹ 13,072.33	₹ 13,242.27	₹ 169.94	₹ 13,072.33	₹ -0.00

Loan Amount	100000
lo.of monthly paymer	15
EMI	-12000
Interest	8%

5. Investment Decision Analysis

Two investment alternatives were evaluated using Net Present Value (NPV). The discount rate used for the analysis was **20%**.

Cash flow patterns for both investments were analysed and NPVs were calculated under different assumptions such as end-of-year, beginning-of-year, and mid-year cash flows.

Results indicated that **Investment 2 produced a higher NPV under end-of-year assumptions**, suggesting it may provide better financial returns.

Decision on Investments

Interest Rate	0.2	
	Cash Flow	
Time	Investement 1	Investement 2
1	-10000	-5000
2	25000	20000
3	-7000	-8000
Total	8000	7000

NPV (End Year)	₹ 5,992.00	₹ 7,000.00
NPV (Beginning Year)	₹ 22,792.62	₹ 19,000.00
NPV (Middle Year)	₹ 28,792.62	₹ 1,000.00

Interest Rate Date	0.2 Cash Flow
15/Jun/2016	5000
14/Oct/2016	5143
30/Apr/2017	8838
10/Nov/2016	-4893
16/Sep/2016	-2134
18/Apr/2017	8047
27/Aug/2016	3908
17/Jul/2016	-4007
NPV	17523.6545

Interest Rate Date	0.2 Cash Flow
15/Mar/2015	0
15/Jun/2016	5000
14/Oct/2016	5143
30/Apr/2017	8838
10/Nov/2016	-4893
16/Sep/2016	-2134
18/Apr/2017	8047
27/Aug/2016	3908
17/Jul/2016	-4007
NPV	13940.18

6. NPV with Irregular Cash Flows

In many real-world scenarios, cash flows occur at irregular intervals. To address this, the NPV of irregular cash flows was calculated using date-based discounting methods.

The analysis produced a total NPV of approximately **₹17,523**, indicating that the investment generates positive value under the given discount rate.

IRR

Cash Flows
10000
-5000
-8500
2000

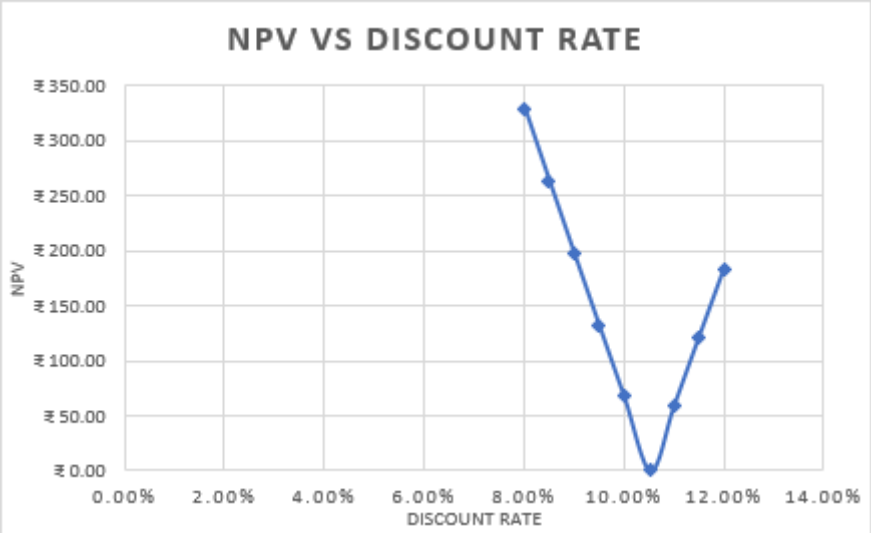
IRR
10.53%

Cash Flows	Guess	IRR
10000		-743.98
-5000	0.05	539.99
-8500	0.15	1087.96
2000	0.2	1584.00
	0.25	2034.59
	0.3	2445.26
	0.35	2820.70
	0.4	3164.95
	0.45	3481.48
	0.5	3773.29
	0.55	-1500.00

7. IRR Calculation

Internal Rate of Return (IRR) represents the discount rate at which the NPV of an investment becomes zero. A trial discount rate method was used to estimate the IRR.

By analysing the NPV at multiple discount rates, the IRR was determined to be approximately **10.53%**.



8. Project Comparison: Project A vs Project B

Two projects were evaluated based on their annual cash flows. Both the IRR and NPV methods were applied to compare their profitability.

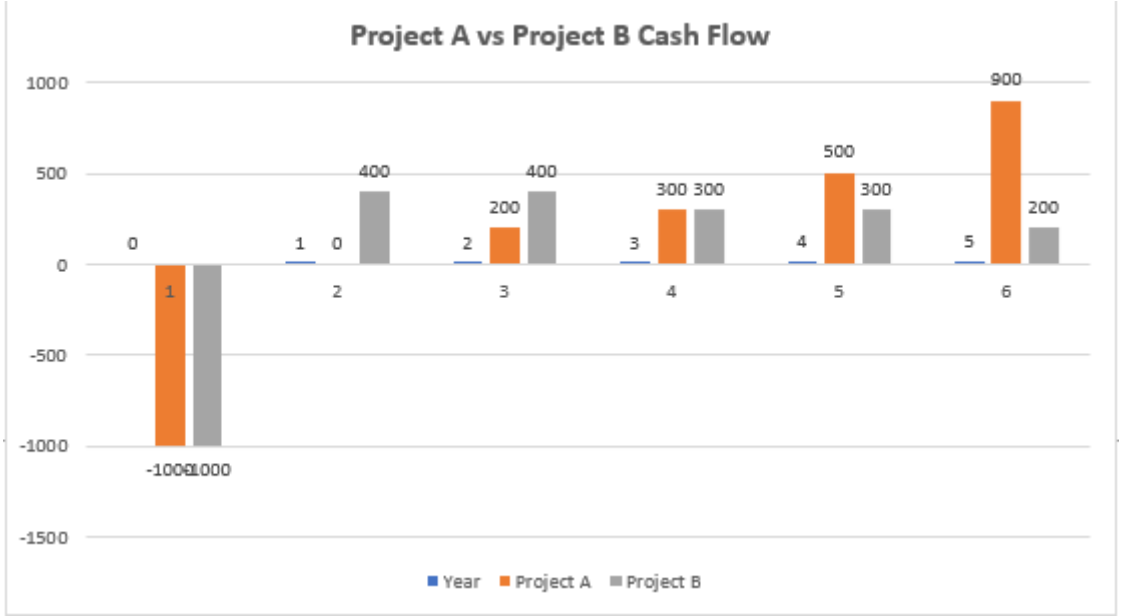
Results:

Project NPV

Project A ₹815.89

Project B ₹552.40

Since **Project A has a higher NPV**, it is considered the better investment option.



9. XIRR Calculation

XIRR was used to calculate the internal rate of return for investments with irregular cash flow dates. Based on the cash flow data and dates provided, the calculated XIRR was **26.42%**, indicating a relatively strong investment performance.

Date	Cash Flow
4/8/2015	-10000
8/15/2015	4000
3/15/2016	3000
4/25/2016	5000
XIRR	26.42%

10. MIRR Analysis

Modified Internal Rate of Return (MIRR) addresses certain limitations of IRR by incorporating both the finance rate and reinvestment rate.

Parameters:

- Finance Rate: 10%
- Reinvestment Rate: 12%

The calculated **MIRR was approximately 6.55%**, providing a more realistic measure of investment profitability.

Discount	NPV
10%	-0.77
25%	0.00
110%	0.89
400%	0.00
500%	-0.21

MIRR	6.55%

Conclusion

This project successfully demonstrated the use of Excel financial functions for evaluating loans and investment decisions. Various analytical tools such as NPV, IRR, XIRR, and MIRR were applied to assess financial performance and compare alternative investment options. The results highlight the importance of discounted cash flow techniques in financial decision-making and illustrate how Excel can be effectively used as a financial analysis tool.